Asset Pricing HW 2

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1 Q1

1.1 Q1(a)

Define X as the payoff matrix:

$$X = \begin{pmatrix} 10 & 10 & 0 \\ 20 & 10 & 10 \\ 30 & 10 & 10 \end{pmatrix},$$

where each column is the payoff of one asset across the three states.

We have complete markets if and only if we can find a unique matrix Ω such that

$$X\Omega = I.$$

Let

$$\Omega = \begin{pmatrix} \omega_{11} & \omega_{12} & \omega_{13} \\ \omega_{21} & \omega_{22} & \omega_{23} \\ \omega_{31} & \omega_{32} & \omega_{33} \end{pmatrix}.$$

Then

$$\begin{pmatrix} 10 & 10 & 0 \\ 20 & 10 & 10 \\ 30 & 10 & 10 \end{pmatrix} \begin{pmatrix} \omega_{11} & \omega_{12} & \omega_{13} \\ \omega_{21} & \omega_{22} & \omega_{23} \\ \omega_{31} & \omega_{32} & \omega_{33} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

For the contingent claim that pays one unit in state 1,

$$10\omega_{11} + 10\omega_{21} = 1,$$

$$20\omega_{11} + 10\omega_{21} + 10\omega_{31} = 0,$$

$$30\omega_{11} + 10\omega_{21} + 10\omega_{31} = 0.$$

Subtracting the second equation from the third gives

$$10\omega_{11} = 0,$$

so

$$\omega_{11} = 0.$$

The first equation then gives

$$\omega_{21} = \frac{1}{10},$$

and the second equation gives

$$\omega_{31} = -\frac{1}{10}.$$

For the contingent claim that pays one unit in state 2,

$$10\omega_{12} + 10\omega_{22} = 0,$$

$$20\omega_{12} + 10\omega_{22} + 10\omega_{32} = 1,$$

$$30\omega_{12} + 10\omega_{22} + 10\omega_{32} = 0.$$

Subtracting the second equation from the third gives

$$10\omega_{12} = -1,$$

so

$$\omega_{12} = -\frac{1}{10}.$$

The first equation gives

$$\omega_{22} = \frac{1}{10},$$

and the second equation gives

$$\omega_{32} = \frac{1}{5}.$$

For the contingent claim that pays one unit in state 3,

$$10\omega_{13} + 10\omega_{23} = 0,$$

$$20\omega_{13} + 10\omega_{23} + 10\omega_{33} = 0,$$

$$30\omega_{13} + 10\omega_{23} + 10\omega_{33} = 1.$$

Subtracting the second equation from the third gives

$$10\omega_{13} = 1,$$

so

$$\omega_{13} = \frac{1}{10}.$$

The first equation gives

$$\omega_{23} = -\frac{1}{10},$$

and the second equation gives

$$\omega_{33} = -\frac{1}{10}.$$

Therefore,

$$\Omega = \begin{pmatrix} 0 & -\frac{1}{10} & \frac{1}{10} \\ \frac{1}{10} & \frac{1}{10} & -\frac{1}{10} \\ -\frac{1}{10} & \frac{1}{5} & -\frac{1}{10} \end{pmatrix}.$$

This matrix is unique and satisfies

$$X\Omega = I.$$

Therefore, the three Arrow–Debreu claims can be replicated using the original assets, so the market is complete.

1.2 Q1(b)

The identity matrix I is the payoff matrix of the three contingent claims. Since

$$X\Omega = I,$$

the columns of Ω give the portfolios of the original assets that replicate the contingent claims.

By the Law of One Price,

$$P_c = \Omega' P,$$

where

$$P = \begin{pmatrix} 16 \\ 8 \\ 6 \end{pmatrix}.$$

Therefore,

$$P_c = \begin{pmatrix} \omega_{11} & \omega_{21} & \omega_{31} \\ \omega_{12} & \omega_{22} & \omega_{32} \\ \omega_{13} & \omega_{23} & \omega_{33} \end{pmatrix} \begin{pmatrix} 16 \\ 8 \\ 6 \end{pmatrix}.$$

Substituting the values of Ω,

$$P_c = \begin{pmatrix} 0 & \frac{1}{10} & -\frac{1}{10} \\ -\frac{1}{10} & \frac{1}{10} & \frac{1}{5} \\ \frac{1}{10} & -\frac{1}{10} & -\frac{1}{10} \end{pmatrix} \begin{pmatrix} 16 \\ 8 \\ 6 \end{pmatrix}.$$

Hence,

$$P_c = \begin{pmatrix} 0(16) + \frac{1}{10}(8) - \frac{1}{10}(6) \\ -\frac{1}{10}(16) + \frac{1}{10}(8) + \frac{1}{5}(6) \\ \frac{1}{10}(16) - \frac{1}{10}(8) - \frac{1}{10}(6) \end{pmatrix} = \begin{pmatrix} \frac{1}{5} \\ \frac{2}{5} \\ \frac{1}{5} \end{pmatrix}.$$

Therefore,

$$\boxed{P_c(1) = \frac{1}{5}, \qquad P_c(2) = \frac{2}{5}, \qquad P_c(3) = \frac{1}{5}.}$$

1.3 Q1(c)

Asset D has payoff vector

$$X_D = \begin{pmatrix} 30 \\ 20 \\ 10 \end{pmatrix}.$$

Equivalently, its payoff can be replicated using the three contingent claims as

$$X_D = I \begin{pmatrix} 30 \\ 20 \\ 10 \end{pmatrix}.$$

By the Law of One Price,

$$P_D = P_c' \begin{pmatrix} 30 \\ 20 \\ 10 \end{pmatrix}.$$

Therefore,

$$P_D = \frac{1}{5}(30) + \frac{2}{5}(20) + \frac{1}{5}(10).$$

Thus,

$$P_D = 6 + 8 + 2 = 16.$$

Therefore,

$$\boxed{P_D = 16.}$$

1.4 Q1(d)

An SDF must price all assets according to

$$P_t = E_t[M_{t+1}X_{t+1}].$$

This must also hold for each Arrow–Debreu claim.

A contingent claim for state n pays one unit in state n and zero otherwise. Therefore,

$$P_c(n) = \pi(n)M(n) \cdot 1.$$

It follows that

$$M(n) = \frac{P_c(n)}{\pi(n)}.$$

Using

$$P_c(1) = \frac{1}{5}, \qquad P_c(2) = \frac{2}{5}, \qquad P_c(3) = \frac{1}{5},$$

and

$$\pi(1) = \frac{1}{4}, \qquad \pi(2) = \frac{1}{2}, \qquad \pi(3) = \frac{1}{4},$$

we obtain

$$M = \begin{pmatrix} \frac{1/5}{1/4} \\ \frac{2/5}{1/2} \\ \frac{1/5}{1/4} \end{pmatrix}.$$

Therefore,

$$M = \begin{pmatrix} \frac{4}{5} \\ \frac{4}{5} \\ \frac{4}{5} \end{pmatrix}.$$

Thus,

$$\boxed{M(1) = M(2) = M(3) = \frac{4}{5}.}$$

The SDF is strictly positive in every state:

$$M(n) > 0 \qquad \text{for every state } n.$$

Suppose an asset has a payoff satisfying

$$X_A(n) \geq 0$$

for every state, with a strictly positive payoff in at least one state. Its price is

$$P_A = E[MX_A].$$

Since $M > 0$,

$$P_A > 0.$$

Therefore, a nonnegative payoff that is positive in at least one state cannot be obtained at a zero or negative price. Hence, there is no arbitrage.

An interesting feature of this economy is that the SDF is constant across states:

$$M(1) = M(2) = M(3).$$

Therefore,

$$\text{Cov}(M, X) = 0$$

for any payoff X. There is no state-dependent risk adjustment coming from the SDF. A dollar received in state 1 is valued in the same way as a dollar received in state 2 or state 3.

1.5 Q1(e)

Using the SDF, the price of asset D is

$$P_D = E[MX_D].$$

Since

$$M = \frac{4}{5} = 0.8$$

in every state,

$$P_D = 0.8 \left[\frac{1}{4}(30) + \frac{1}{2}(20) + \frac{1}{4}(10) \right].$$

Therefore,

$$P_D = 0.8 \left(7.5 + 10 + 2.5 \right).$$

Hence,

$$P_D = 0.8(20) = 16.$$

Therefore,

$$\boxed{P_D = 16,}$$

which is the same price as the one obtained in Q1(c).

2 Q2

2.1 Q2(a)

Using the 10 value-weighted book-to-market decile portfolios, monthly excess returns are constructed as

$$xR_{j,t} = R_{j,t} - RF_t, \qquad j = 1, \ldots, 10.$$

Let

$$\hat{\mu} = \hat{E}[xR_t], \qquad \hat{\Sigma} = \widehat{\mathrm{Var}}(xR_t).$$

The coefficients of the maximum Sharpe ratio portfolio are

$$\hat{b} = \hat{\Sigma}^{-1}\hat{\mu},$$

and the excess return of the maximum Sharpe ratio portfolio is

$$xR_t^* = \hat{b}' xR_t.$$

Using the normalization

$$\hat{E}[M_t^*] = 1,$$

the SDF is constructed as

$$\hat{M}_t^* = \hat{a} - \hat{b}' xR_t, \qquad \hat{a} = 1 + \hat{b}'\hat{\mu}.$$

The estimate of the constant is

$$\hat{a} = 1.0368,$$

and the sample pricing condition

$$\hat{E}[\hat{M}_t^* xR_t] \approx 0$$

is satisfied.

Figure 1 plots the estimated $\hat{b}$ coefficients across the 10 book-to-market deciles.

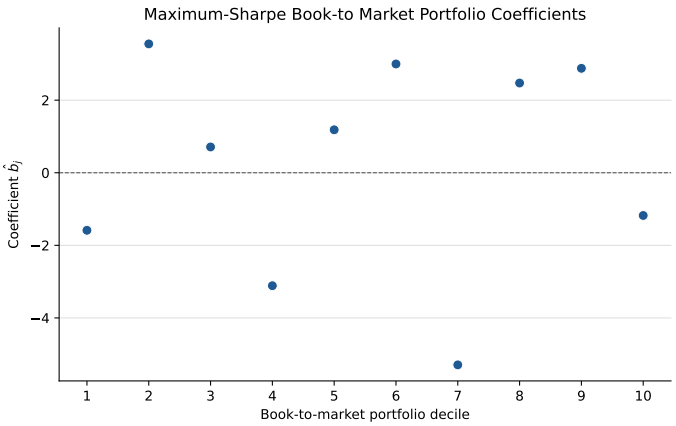


Figure 1: Estimated b coefficients across book-to-market decile portfolios

The estimated coefficients vary substantially across the book-to-market deciles and are not monotonic. This pattern arises because

$$\hat{b} = \hat{\Sigma}^{-1}\hat{\mu},$$

so each coefficient depends not only on the average excess return of that portfolio, but also on its covariance with the other portfolios. Therefore, the coefficients do not simply increase or decrease with the book-to-market decile.

2.2 Q2(b)

The analysis in Q2(a) is repeated using the 10 value-weighted momentum decile portfolios.

The estimate of the constant is

$$\hat{a} = 1.0666,$$

and the sample pricing condition

$$\hat{E}[\hat{M}_t^* x R_t] \approx 0$$

is satisfied.

Figure 2 plots the estimated $\hat{b}$ coefficients across the 10 momentum deciles.

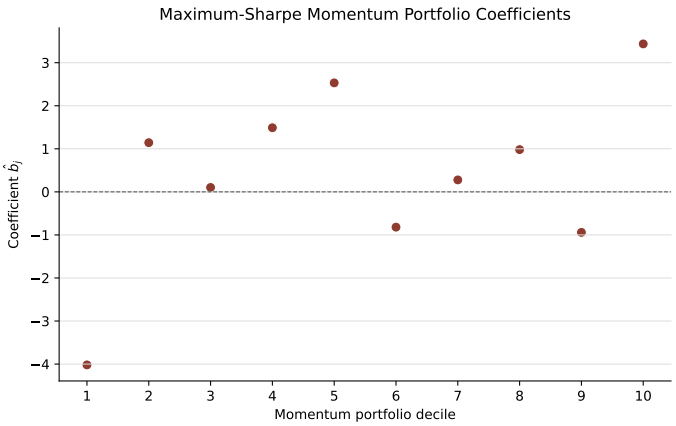


Figure 2: Estimated b coefficients across momentum decile portfolios

The estimated coefficients vary substantially across the momentum deciles and are not monotonic. As in Q2(a), this reflects both the average excess returns and the covariance structure across the portfolios.

The correlation between the SDF from Q2(a) and the SDF from Q2(b) is

$$\text{Corr}\left(M^*_{BM}, M^*_{MOM}\right) = 0.3555.$$

The positive correlation indicates that the two SDFs share some common variation, but the relatively low correlation suggests that the book-to-market and momentum portfolios contain different information about the maximum Sharpe ratio direction.

2.3 Q2(c)

The 10 book-to-market portfolios and 10 momentum portfolios are combined to construct the SDF using 20 excess returns.

The estimate of the constant is

$$\hat{a} = 1.0927.$$

The sample pricing condition

$$\hat{E}[\hat{M}^*_t x R_t] \approx 0$$

is satisfied for all 20 portfolios.

Figure 3 plots the estimated $\hat{b}$ coefficients for the book-to-market and momentum portfolios.

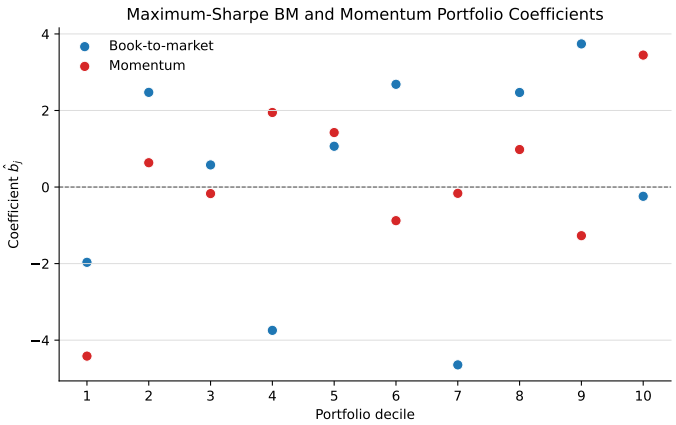


Figure 3: Estimated b coefficients for book-to-market and momentum portfolios

The coefficients vary substantially across both sets of portfolios and are not monotonic. This is because the coefficients depend on both the average excess returns and the covariance structure across all 20 portfolios.

The monthly Sharpe ratios of the maximum Sharpe ratio portfolios are

$$SR_{BM} = 0.1919, \qquad SR_{MOM} = 0.2581, \qquad SR_{BM+MOM} = 0.3045.$$

The maximum Sharpe ratio increases when the BM and momentum portfolios are combined because the larger asset span provides additional portfolio opportunities. This is consistent with the Hansen-Jagannathan property that the maximum attainable Sharpe ratio places a lower bound on SDF volatility.

2.4 Q2(d)

For each of the 20 portfolios, the pricing error is calculated under the SDFs from Q2(a), Q2(b), and Q2(c) as

$$\hat{\alpha}_p = \hat{E}[xR_p] - \widehat{\mathrm{Cov}}(-M^*, xR_p).$$

The SDF from Q2(a) prices the 10 book-to-market portfolios almost exactly in sample, but produces larger pricing errors for the momentum portfolios. The SDF from Q2(b) prices the momentum portfolios almost exactly but produces pricing errors for the book-to-market portfolios.

The SDF from Q2(c) prices all 20 portfolios almost exactly, with pricing errors close to zero. This is because the Q2(c) SDF is constructed using both the book-to-market and momentum portfolios, while the SDFs in Q2(a) and Q2(b) are constructed using only one set of portfolios.

Table 1: Pricing errors under the Q2(a), Q2(b), and Q2(c) SDFs.

Portfolio	$\widehat{\alpha}^{Q2(a)}$	$\widehat{\alpha}^{Q2(b)}$	$\widehat{\alpha}^{Q2(c)}$
BM: Lo 10	0.000000	-0.001511	0.000000
BM: 2-Dec	0.000000	-0.000101	0.000000
BM: 3-Dec	0.000000	-0.000223	0.000000
BM: 4-Dec	0.000000	-0.000778	0.000000
BM: 5-Dec	0.000000	0.000504	0.000000
BM: 6-Dec	0.000000	0.001261	0.000000
BM: 7-Dec	0.000000	0.000281	0.000000
BM: 8-Dec	0.000000	0.002376	0.000000
BM: 9-Dec	0.000000	0.003450	0.000000
BM: Hi 10	0.000000	0.003198	0.000000
MOM: Lo PRIOR	-0.010157	0.000000	0.000000
MOM: PRIOR 2	-0.005051	0.000000	0.000000
MOM: PRIOR 3	-0.003272	0.000000	0.000000
MOM: PRIOR 4	-0.001488	0.000000	0.000000
MOM: PRIOR 5	-0.000882	0.000000	0.000000
MOM: PRIOR 6	-0.000263	0.000000	0.000000
MOM: PRIOR 7	0.000509	0.000000	0.000000
MOM: PRIOR 8	0.001902	0.000000	0.000000
MOM: PRIOR 9	0.002240	0.000000	0.000000
MOM: Hi PRIOR	0.005418	0.000000	0.000000

2.5 Q2(e)

The SDF coefficients are estimated using the sample from July 1927 to December 1989. These coefficients are then held fixed and used to construct the out-of-sample SDF from January 1990 onward.

For each of the 20 portfolios, the out-of-sample pricing error is calculated as

$$\hat{\alpha}_p^{OS} = \hat{E}_{OS}[xR_p] - \widehat{\mathrm{Cov}}_{OS}(-M_{OS}^*, xR_p).$$

The out-of-sample pricing errors are much larger than the in-sample pricing errors in Q2(d). The average absolute pricing error is about 0.81% per month for the book-to-market portfolios and 1.11% per month for the momentum portfolios.

Overall, the SDF estimated using pre-1990 data does not price the 20 portfolios as well after 1990. This suggests that the relation between the portfolio returns and the estimated SDF is not stable across the two sample periods.

Table 2: Out-of-sample pricing errors for the Q2(e) SDF.

Portfolio	$\widehat{\alpha}_{OS}$
BM: Lo 10	0.010971
BM: 2-Dec	0.007858
BM: 3-Dec	0.009002
BM: 4-Dec	0.006957
BM: 5-Dec	0.007986
BM: 6-Dec	0.007178
BM: 7-Dec	0.006977
BM: 8-Dec	0.005923
BM: 9-Dec	0.008269
BM: Hi 10	0.009709
MOM: Lo PRIOR	0.027055
MOM: PRIOR 2	0.016483
MOM: PRIOR 3	0.013905
MOM: PRIOR 4	0.012077
MOM: PRIOR 5	0.009180
MOM: PRIOR 6	0.008297
MOM: PRIOR 7	0.007601
MOM: PRIOR 8	0.006444
MOM: PRIOR 9	0.004322
MOM: Hi PRIOR	0.005453

3 Q3

3.1 Q3(a)

The investor solves

$$\max_{\omega_t} E_t[U(W_{t+1})]$$

subject to

$$W_{t+1} = W_t \omega_t' R_{t+1}$$

and

$$\mathbf{1}'\omega_t = 1.$$

The Lagrangian is

$$\mathcal{L} = E_t\left[U\left(W_t\omega_t' R_{t+1}\right)\right] + \lambda_t\left(1 - \mathbf{1}'\omega_t\right).$$

Taking the derivative with respect to ω_t, the first-order condition is

$$\frac{\partial \mathcal{L}}{\partial \omega_t} = E_t\left[U'(W_{t+1})W_t R_{t+1}\right] - \lambda_t \mathbf{1} = 0.$$

Therefore, for any two assets j and k,

$$E_t\left[U'(W_{t+1})W_t R_{j,t+1}\right] = E_t\left[U'(W_{t+1})W_t R_{k,t+1}\right].$$

Subtracting the right-hand side from the left-hand side gives

$$E_t\left[U'(W_{t+1})W_t\left(R_{j,t+1} - R_{k,t+1}\right)\right] = 0.$$

Let asset k be the baseline asset $R_{f,t+1}$. Then

$$R_{j,t+1} - R_{f,t+1} = xR_{j,t+1}.$$

Therefore,

$$E_t\left[U'(W_{t+1})W_t x R_{j,t+1}\right] = 0.$$

Under CRRA utility,

$$U(W) = \frac{W^{1-\gamma}}{1-\gamma}.$$

Therefore, marginal utility is

$$U'(W) = W^{-\gamma}.$$

Hence,

$$U'(W_{t+1}) = W_{t+1}^{-\gamma}.$$

Since

$$W_{t+1} = W_t R_{w,t+1},$$

we have

$$U'(W_{t+1}) = \left(W_t R_{w,t+1}\right)^{-\gamma}.$$

Thus,

$$U'(W_{t+1}) = W_t^{-\gamma} R_{w,t+1}^{-\gamma}.$$

Substituting this into the first-order condition gives

$$E_t\left[W_t^{-\gamma} R_{w,t+1}^{-\gamma} W_t x R_{j,t+1}\right] = 0.$$

Combining the powers of W_t,

$$E_t\left[W_t^{1-\gamma} R_{w,t+1}^{-\gamma} x R_{j,t+1}\right] = 0.$$

Since W_t is known at time t, $W_t^{1-\gamma}$ can be taken outside the conditional expectation:

$$W_t^{1-\gamma} E_t \left[R_{w,t+1}^{-\gamma} x R_{j,t+1} \right] = 0.$$

Since

$$W_t^{1-\gamma} \neq 0,$$

it follows that

$$E_t \left[R_{w,t+1}^{-\gamma} x R_{j,t+1} \right] = 0.$$

This holds for every asset j. Therefore, in vector form,

$$\boxed{E_t \left[R_{w,t+1}^{-\gamma} x R_{t+1} \right] = 0.}$$

Thus, the SDF

$$\boxed{M_{t+1} = R_{w,t+1}^{-\gamma}}$$

prices the vector of excess returns.

3.2 Q3(b)

Using the unconditional SDF moment condition

$$E \left[R_{w,t}^{-\gamma} x R_{w,t} \right] = 0,$$

with

$$R_{w,t} = e^{re_t}, \qquad R_{f,t} = e^{rf_t}, \qquad xR_{w,t} = R_{w,t} - R_{f,t},$$

γ is estimated by GMM using $T = 1129$ monthly observations.
The estimate is

$$\hat{\gamma} = 1.9061.$$

The standard error is calculated using a Newey-West (1987, 1994) HAC estimator with a Bartlett kernel and a data-driven bandwidth. The selected bandwidth is

$$L = 23,$$

which gives

$$SE(\hat{\gamma}) = 0.5941.$$

3.3 Q3(c)

Using the two unconditional SDF moment conditions

$$E\left[R_{w,t}^{-\gamma} x R_{w,t}\right] = 0$$

and

$$E\left[R_{w,t}^{-\gamma}\left(z_{t-1} x R_{w,t}\right)\right] = 0,$$

γ is estimated by GMM using an identity weighting matrix.

The estimation uses $T = 973$ monthly observations. The data-driven Newey-West procedure selects

$$L = 22.$$

The estimate of γ, its Newey-West standard error, the pricing errors and corresponding t-statistics, and the joint chi-square test are reported below.

Table 3: Identity-weighted GMM estimates and pricing errors for Q3(c).

Quantity	Estimate	Standard error	t-statistic
$\widehat{\gamma}$	2.985082	0.714967	
Wealth-portfolio pricing error	-0.003118	0.003885	-0.802
Managed-portfolio pricing error	0.002229	0.002777	0.802
Joint $\chi^2(1)$	0.643863	p-value	0.422315
Observations	973		

The joint test does not reject the null hypothesis that the two pricing errors are jointly zero. Therefore, the CRRA SDF is broadly consistent with both the wealth-portfolio and managed-portfolio moment conditions.

3.4 Q3(d)

Using the same two moment conditions as in Q3(c), γ is re-estimated using two-step GMM.

In the first step, the identity weighting matrix is used. In the second step, the inverse of the Newey-West long-run covariance matrix from the first step is used as the weighting matrix.

The estimation uses the same $T = 973$ monthly observations as in Q3(c). The data-driven Newey-West procedure selects

$$L = 22.$$

The two-step GMM estimate of γ, its Newey-West standard error, the pricing errors and corresponding t-statistics, and the joint chi-square test are reported below.

The joint test again does not reject the null hypothesis that the two pricing errors are jointly zero. Therefore, the two-step GMM specification remains consistent with the wealth-portfolio and managed-portfolio moment conditions.

Table 4: Efficient two-step GMM estimates and pricing errors for Q3(d).

Quantity	Estimate	Standard error	t-statistic
$\widehat{\gamma}_{2step}$	2.944766	0.716937	
Wealth-portfolio pricing error	-0.001821	0.002270	-0.802
Managed-portfolio pricing error	0.004043	0.005038	0.802
Joint $\chi^2(1)$	0.643958	p-value	0.422281
Observations	973		

3.5 Q3(e)

Using the wealth portfolio together with the 10 book-to-market and 10 momentum decile portfolios, γ is estimated by GMM using 21 unconditional moment conditions.

Using $T = 1129$ monthly observations, the GMM estimate is

$$\hat{\gamma} = 1.9083,$$

with a Newey-West standard error of

$$SE(\hat{\gamma}) = 0.5863.$$

The data-driven Newey-West procedure selects

$$L = 23.$$

The pricing errors and corresponding t-statistics for the wealth portfolio, the 10 book-to-market portfolios, and the 10 momentum portfolios are reported below.

The joint test that all 21 pricing errors are zero gives

$$\chi^2(20) = 92.1354, \qquad p < 0.0001.$$

Thus, the null hypothesis that all pricing errors are jointly zero is rejected. This suggests that the CRRA SDF is not able to jointly price the wealth, book-to-market, and momentum portfolios, with the strongest evidence of mispricing coming from the momentum portfolios.

Table 5: Pricing errors for the Q3(e) identity-weighted GMM model.

Portfolio	Pricing error	Standard error	t-statistic
Wealth portfolio (xR_w)	-0.000108	0.002904	-0.037
BM Lo 10	-0.001137	0.001284	-0.885
BM 2-Dec	0.000604	0.001060	0.570
BM 3-Dec	0.000542	0.001443	0.375
BM 4-Dec	-0.000669	0.001380	-0.485
BM 5-Dec	0.000588	0.001557	0.378
BM 6-Dec	0.000769	0.001665	0.462
BM 7-Dec	-0.000233	0.002005	-0.116
BM 8-Dec	0.001699	0.002066	0.822
BM 9-Dec	0.003217	0.002796	1.151
BM Hi 10	0.001712	0.003112	0.550
MOM Lo PRIOR	-0.006566	0.003240	-2.026
MOM PRIOR 2	-0.002502	0.002488	-1.006
MOM PRIOR 3	-0.001939	0.002186	-0.887
MOM PRIOR 4	-0.001148	0.001864	-0.616
MOM PRIOR 5	-0.000097	0.001533	-0.063
MOM PRIOR 6	0.000111	0.001519	0.073
MOM PRIOR 7	0.000546	0.001283	0.426
MOM PRIOR 8	0.001469	0.001030	1.426
MOM PRIOR 9	0.001904	0.001266	1.503
MOM Hi PRIOR	0.004200	0.001377	3.051

4 Q4

4.1 Q4(a)

From Q3(a), the nonlinear SDF is

$$M_{t+1} = R_{w,t+1}^{-\gamma}.$$

Let

$$\mu_t = E_t[R_{w,t+1}].$$

Define

$$f(R) = R^{-\gamma}.$$

Then

$$f'(R) = -\gamma R^{-\gamma-1}.$$

Using a first-order Taylor expansion of $R_{w,t+1}^{-\gamma}$ around μ_t,

$$f(R_{w,t+1}) \approx f(\mu_t) + f'(\mu_t)\left(R_{w,t+1} - \mu_t\right).$$

Therefore,

$$M_{t+1} = R_{w,t+1}^{-\gamma}$$

can be approximated by

$$M_{t+1} \approx \mu_t^{-\gamma} - \gamma \mu_t^{-\gamma-1} \left(R_{w,t+1} - \mu_t \right).$$

Expanding the last term,

$$M_{t+1} \approx \mu_t^{-\gamma} - \gamma \mu_t^{-\gamma-1} R_{w,t+1} + \gamma \mu_t^{-\gamma-1} \mu_t.$$

Since

$$\mu_t^{-\gamma-1} \mu_t = \mu_t^{-\gamma},$$

we obtain

$$M_{t+1} \approx \mu_t^{-\gamma} + \gamma \mu_t^{-\gamma} - \gamma \mu_t^{-\gamma-1} R_{w,t+1}.$$

Therefore,

$$M_{t+1} \approx (1+\gamma) \mu_t^{-\gamma} - \gamma \mu_t^{-\gamma-1} R_{w,t+1}.$$

Comparing this expression with

$$M_{t+1} \approx a_t - b_t R_{w,t+1},$$

we have

$$\boxed{a_t = (1+\gamma) \mu_t^{-\gamma}}$$

and

$$\boxed{b_t = \gamma \mu_t^{-\gamma-1}.}$$

Since

$$\mu_t = E_t[R_{w,t+1}],$$

these can equivalently be written as

$$\boxed{a_t = (1+\gamma) \left(E_t[R_{w,t+1}] \right)^{-\gamma}}$$

and

$$\boxed{b_t = \gamma \left(E_t[R_{w,t+1}] \right)^{-\gamma-1}.}$$

Therefore,

$$\boxed{M_{t+1} \approx a_t - b_t R_{w,t+1}.}$$

The advantage of this linear SDF is that it is linear in $R_{w,t+1}$, which leads to a simpler GMM estimator.

One disadvantage is that the linear SDF can become negative when $R_{w,t+1}$ is sufficiently large:

$$a_t - b_t R_{w,t+1} < 0.$$

In contrast, the original power-utility SDF

$$M_{t+1} = R_{w,t+1}^{-\gamma}$$

is always positive as long as

$$R_{w,t+1} > 0.$$

4.2 Q4(b)

Using the normalized linear SDF

$$M_t = 1 - b\left(R_{w,t} - E[R_w]\right),$$

b is estimated by GMM using the wealth portfolio together with the 10 book-to-market and 10 momentum decile portfolios, for a total of 21 unconditional moment conditions. An identity weighting matrix is used.

Using $T = 1129$ monthly observations from December 1927 to December 2021, the GMM estimate is

$$\hat{b} = 2.0141,$$

with a Newey-West standard error of

$$SE(\hat{b}) = 0.6001.$$

The data-driven Newey-West procedure selects

$$L = 23.$$

The pricing errors and corresponding t-statistics for the wealth portfolio, the 10 book-to-market portfolios, and the 10 momentum portfolios are reported below.

The joint test that all 21 pricing errors are zero gives

$$\chi^2(20) = 103.9770, \qquad p = 2.43 \times 10^{-13}.$$

Thus, the null hypothesis that all pricing errors are jointly zero is strongly rejected. This suggests that the linear SDF is not able to jointly price the wealth, book-to-market, and momentum portfolios.

Table 6: Identity-weighted GMM results for the normalized linear SDF.

Quantity or portfolio	Estimate	t-statistic
$\widehat{b}$	2.014081	
Newey-West standard error of $\widehat{b}$	0.600057	
Observations	1129	
Common bandwidth L	23	
Joint $\chi^2(20)$	103.976951	
Joint-test p-value	2.430e-13	
Pricing errors		
Wealth portfolio (xR_w)	-0.000808	-0.359
BM Lo 10	-0.000379	-0.285
BM 2-Dec	0.001451	1.303
BM 3-Dec	0.000991	0.767
BM 4-Dec	-0.000164	-0.127
BM 5-Dec	0.000801	0.577
BM 6-Dec	0.001224	0.843
BM 7-Dec	-0.000437	-0.257
BM 8-Dec	0.001494	0.892
BM 9-Dec	0.002587	1.183
BM Hi 10	0.001103	0.411
MOM Lo PRIOR	-0.007001	-2.394
MOM PRIOR 2	-0.001644	-0.736
MOM PRIOR 3	-0.000851	-0.411
MOM PRIOR 4	-0.000153	-0.088
MOM PRIOR 5	0.000850	0.614
MOM PRIOR 6	0.000788	0.581
MOM PRIOR 7	0.000948	0.751
MOM PRIOR 8	0.001821	1.866
MOM PRIOR 9	0.002194	1.833
MOM Hi PRIOR	0.004591	3.271

4.3 Q4(c)

Using the same 21 moment conditions as in Q4(b), b is re-estimated using efficient GMM. The inverse of the Newey-West long-run covariance matrix from Q4(b) is used as the weighting matrix.

The sample moment condition can be written as

$$\bar{g}(b) = \hat{\mu} - b\hat{c},$$

where

$$\hat{\mu} = \hat{E}[xR_t]$$

and

$$\hat{c} = \hat{E}\left[(R_{w,t} - \bar{R}_w)xR_t\right].$$

Using the efficient weighting matrix

$$W = \hat{S}^{-1},$$

the GMM objective function is

$$Q(b) = (\hat{\mu} - b\hat{c})'\hat{S}^{-1}(\hat{\mu} - b\hat{c}).$$

Taking the derivative with respect to b,

$$\frac{\partial Q(b)}{\partial b} = -2\hat{c}'\hat{S}^{-1}(\hat{\mu} - b\hat{c}).$$

Setting the first-order condition equal to zero gives

$$\hat{c}'\hat{S}^{-1}(\hat{\mu} - \hat{b}_{\text{eff}}\hat{c}) = 0.$$

Therefore, the efficient GMM estimator has the analytical expression

$$\hat{b}_{\text{eff}} = \frac{\hat{c}'\hat{S}^{-1}\hat{\mu}}{\hat{c}'\hat{S}^{-1}\hat{c}}.$$

Using $T = 1129$ monthly observations, the efficient GMM estimate is

$$\hat{b}_{\text{eff}} = 2.5984,$$

with a Newey-West standard error of

$$SE(\hat{b}_{\text{eff}}) = 0.4582.$$

The pricing errors and corresponding t-statistics for the wealth portfolio, the 10 book-to-market portfolios, and the 10 momentum portfolios are reported below.

The joint test that all 21 pricing errors are zero gives

Table 7: Efficient GMM results for the normalized linear SDF.

Quantity or portfolio	Estimate	t-statistic
$\widehat{b}_{eff}$	2.598366	
Newey-West standard error of $\widehat{b}_{eff}$	0.458231	
Observations	1129	
Q4(b) common bandwidth L	23	
Joint $\chi^2(20)$	103.976951	
Joint-test p-value	2.430e-13	
Pricing errors		
Wealth portfolio (xR_w)	-0.025407	-1.548
BM Lo 10	-0.002267	-1.133
BM 2-Dec	-0.000264	-0.157
BM 3-Dec	-0.000675	-0.427
BM 4-Dec	-0.002179	-1.053
BM 5-Dec	-0.001080	-0.521
BM 6-Dec	-0.000764	-0.342
BM 7-Dec	-0.002561	-1.077
BM 8-Dec	-0.000761	-0.359
BM 9-Dec	0.000294	0.135
BM Hi 10	-0.001595	-0.609
MOM Lo PRIOR	-0.009314	-2.740
MOM PRIOR 2	-0.003518	-1.342
MOM PRIOR 3	-0.002566	-0.962
MOM PRIOR 4	-0.002019	-0.855
MOM PRIOR 5	-0.000831	-0.459
MOM PRIOR 6	-0.000981	-0.510
MOM PRIOR 7	-0.000963	-0.494
MOM PRIOR 8	-0.000124	-0.077
MOM PRIOR 9	0.000170	0.098
MOM Hi PRIOR	0.002436	1.320

$$\chi^2(20) = 103.9770, \qquad p = 2.43 \times 10^{-13}.$$

Thus, the null hypothesis that all pricing errors are jointly zero is strongly rejected. This suggests that the efficiently estimated linear SDF is not able to jointly price the wealth, book-to-market, and momentum portfolios.

4.4 Q4(d)

The linear SDF

$$M_t = a - bR_{w,t}$$

implies a linear beta-pricing model. For each of the 21 assets, the time-series regression

$$xR_{j,t} = \alpha_j + \beta_j R_{w,t} + \epsilon_{j,t}$$

is estimated, where α_j measures the pricing error.

The factor risk premium is estimated from a time-series regression of $R_{w,t}$ on a constant. Using $T = 1129$ monthly observations, the estimate is

$$\hat{\lambda} = 1.0877,$$

with a Newey-West standard error of

$$SE(\hat{\lambda}) = 0.0180.$$

The data-driven Newey-West procedure selects

$$L = 23.$$

The estimated pricing errors and corresponding Newey-West t-statistics for the wealth portfolio, the 10 book-to-market portfolios, and the 10 momentum portfolios are reported below.

The GRS test of the null hypothesis

$$H_0 : \alpha_1 = \alpha_2 = \cdots = \alpha_{21} = 0$$

gives

$$GRS = 1319.6632, \qquad p < 0.0001.$$

Thus, the null hypothesis that all pricing errors are jointly zero is strongly rejected. This suggests that the linear beta-pricing model does not jointly price the wealth, book-to-market, and momentum portfolios.

Table 8: Time-series beta-pricing results for the linear SDF model.

Quantity or portfolio	Estimate	t-statistic
$\widehat{\lambda}$	1.087734	
Newey-West standard error of $\widehat{\lambda}$	0.017994	
Lambda bandwidth L	23	
Observations	1129	
GRS $F(21, 1107)$	1319.663199	
GRS p-value	2.391×10^{-764}	
Time-series regression alphas		
Wealth portfolio (xR_w)	-0.999425	-30.945
BM Lo 10	-0.077025	-7.751
BM 2-Dec	-0.068177	-7.956
BM 3-Dec	-0.066631	-7.254
BM 4-Dec	-0.081969	-6.568
BM 5-Dec	-0.075541	-6.313
BM 6-Dec	-0.079485	-6.256
BM 7-Dec	-0.086628	-6.217
BM 8-Dec	-0.090069	-6.152
BM 9-Dec	-0.090472	-5.705
BM Hi 10	-0.108431	-5.450
MOM Lo PRIOR	-0.100904	-4.308
MOM PRIOR 2	-0.077742	-4.789
MOM PRIOR 3	-0.070453	-4.949
MOM PRIOR 4	-0.075904	-5.695
MOM PRIOR 5	-0.067373	-6.425
MOM PRIOR 6	-0.071037	-6.241
MOM PRIOR 7	-0.076635	-7.300
MOM PRIOR 8	-0.077156	-8.411
MOM PRIOR 9	-0.079958	-8.302
MOM Hi PRIOR	-0.082897	-8.969

4.5 Q4(e)

The linear beta-pricing model is estimated using Fama–MacBeth regressions. First, the factor loadings β_j are estimated for the 21 assets. Then, a cross-sectional regression is run in each month to obtain the factor risk premium λ_t and the corresponding pricing errors $\alpha_{j,t}$.

The Fama–MacBeth estimate of the factor risk premium is

$$\hat{\lambda} = \frac{1}{T}\sum_{t=1}^{T}\hat{\lambda}_t = 0.0851.$$

The standard error is calculated from a time-series regression of $\hat{\lambda}_t$ on a constant using Newey-West standard errors. The resulting standard error is

$$SE(\hat{\lambda}) = 0.0186.$$

The data-driven Newey-West procedure selects

$$L = 23,$$

using $T = 1129$ monthly observations.

For each asset j, the average pricing error is

$$\hat{\alpha}_j = \frac{1}{T}\sum_{t=1}^{T}\hat{\alpha}_{j,t},$$

and its t-statistic is calculated using Newey-West standard errors from a time-series regression of $\hat{\alpha}_{j,t}$ on a constant.

The factor risk premium estimate and the pricing errors and corresponding t-statistics for the wealth portfolio, the 10 book-to-market portfolios, and the 10 momentum portfolios are reported below.

Most pricing errors are not statistically significant. However, the lowest momentum portfolio has a significantly negative pricing error, while the highest momentum portfolio has a significantly positive pricing error. This suggests that the linear beta-pricing model does not fully capture the return spread across the momentum portfolios.

Table 9: Fama–MacBeth results for the linear beta-pricing model.

Quantity or portfolio	Estimate	t-statistic
$\widehat{\lambda}$	0.085133	
Newey-West standard error of $\widehat{\lambda}$	0.018579	
Lambda bandwidth L	23	
Observations	1129	
Average cross-sectional pricing errors		
Wealth portfolio (xR_w)	-0.000808	-0.444
BM Lo 10	-0.000379	-0.313
BM 2-Dec	0.001451	1.484
BM 3-Dec	0.000991	0.871
BM 4-Dec	-0.000164	-0.143
BM 5-Dec	0.000801	0.659
BM 6-Dec	0.001224	1.192
BM 7-Dec	-0.000437	-0.342
BM 8-Dec	0.001494	1.115
BM 9-Dec	0.002587	1.502
BM Hi 10	0.001103	0.490
MOM Lo PRIOR	-0.007001	-2.781
MOM PRIOR 2	-0.001644	-0.832
MOM PRIOR 3	-0.000851	-0.550
MOM PRIOR 4	-0.000153	-0.106
MOM PRIOR 5	0.000850	0.749
MOM PRIOR 6	0.000788	0.649
MOM PRIOR 7	0.000948	0.925
MOM PRIOR 8	0.001821	1.905
MOM PRIOR 9	0.002194	1.822
MOM Hi PRIOR	0.004591	3.151